



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

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**INDIAN INSTITUTE OF INFORMATION
TECHNOLOGY SURAT-394190**

Blood Group Detection Using InfraRed Image and Machine Learning

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1. Problem Statement

Manual blood group detection is time-consuming, error-prone, and inaccessible in remote areas.

High demand for rapid, reliable, and automated solutions.

1 Significance

Faster blood group identification is crucial in emergency and medical scenarios.

2 Benefits

Reduces human errors and improves the accuracy of blood group testing.

Abstract

This project focuses on automating blood group detection using spectroscopic images. Our approach utilizes a Machine learning model based on the VGG16 architecture, specifically fine-tuned for this classification task. This model has achieved a high accuracy of 87.34%, demonstrating its potential for efficient and reliable blood group determination. We have also developed a user-friendly web application that allows for easy integration into various medical settings.

Objective

Develop an automated system for blood group detection using image analysis and Machine learning.

Approach

Employ a Machine learning model based on VGG16 architecture, fine-tuned for 8-class blood group classification.

Results

Achieved a 87.34% accuracy on the test dataset and developed a user-friendly web application for real-time predictions.

2. Objective

To develop an automated blood group detection system using spectroscopic images and machine learning.

Improve speed, reliability, and accessibility.

This project explores the use of machine learning to automate blood group detection using spectroscopic image processing. The goal is to develop a system that can accurately predict blood groups from images, leading to faster and more reliable blood group identification.



Traditional Method Error (Not 100% Accuracy)

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Authority Signature

Class	Blood Group
8 th	AB+
After 8 th	O-



3. Proposed Solution

VGG16-based ML model
trained on spectroscopic
images

Flask-based web app for real-
real-time detection

87.34% test accuracy



Data Processing

Methodology

The methodology involves a series of steps, starting with data Preprocessing , followed by model architecture, evaluation, and finally, deployment. Each step plays a crucial role in building a robust and accurate system for automated blood group detection.

Data Preprocessing

Image resizing, augmentation, and splitting into training, validation, and test sets.

Evaluation

Assessment of model performance using metrics such as accuracy, confusion matrix, and loss.

1

2

3

4

Model Architecture

Utilization of a pretrained VGG16 network, fine-tuned for 8-class classification, with custom dense layers and softmax activation.

Deployment

Implementation of a Flask-based web application for real-time blood group predictions.

4. Market Research & Validation

118M blood donations per year globally (WHO)

Hospitals and emergency services need faster diagnostics

Online survey: 86% doctors support AI-aided diagnostics

5. Target Market

Hospitals & Blood Banks

Emergency Services

Remote Clinics

Military & Disaster Teams

Medical Research Labs

6. Competitive Analysis

Method	Time	Accuracy	Portability	Human Error
Traditional Lab	High	High	Low	High
BioVision (ML)	Low	High	High	Low

7. Business Model

Revenue Streams (How We Earn)

#	Revenue Stream	Model	Estimated Revenue (Annual)
1	B2B Licensing to Hospitals	₹1,50,000/year per hospital; target 10 hospitals in year 1	₹15,00,000
2	Subscription for Research Labs	₹3,000/month per lab; targeting 20 labs	₹7,20,000
3	Freemium Web App (Premium Users)	₹499/month per premium user; assume 1,000 paid users	₹59,88,000
4	Infrared Imaging Device (Hardware)	₹15,000/device; expected sale of 500 devices/year	₹75,00,000
	Total Potential Revenue (Year 1)		₹1,57,08,000

Cost Structure (Where We Spend)

Category	Description	Estimated Annual Cost
Cloud Hosting	AWS/GCP server usage, storage, and bandwidth for API and web deployment	₹2,00,000
Model Maintenance & Updates	Developer salaries (part-time), dataset cleaning, retraining, bug fixes	₹4,00,000
Customer Support & Management	Handling client issues, onboarding support, and technical documentation	₹1,00,000
Hardware Manufacturing (IR device)	Manufacturing cost per unit (₹8,000 x 500 units)	₹40,00,000
Marketing & Branding	Digital ads, conference demos, branding materials (posters, logos, social media)	₹2,50,000
Legal & Compliance	Licensing, data privacy compliance (e.g., HIPAA equivalent), consulting fees	₹1,00,000
	Total Estimated Cost (Year 1)	50

 **Projected Year-1 Profit:**
Revenue (₹1,57,08,000) – Cost (₹50,50,000) = ₹1,06,58,000

9. Dataset Overview



Source: Kaggle



Classes: A+, A-, AB+, AB-, B+, B+, B-, O+



Samples: Train (46), Validation (14), Test (11)

Dataset Overview	Details
Number of Classes	7(A+, A-, AB+, AB-, B+,B-,O+)
Training Samples	46
Validation Samples	14
Test Samples	11

9. Updated Dataset Overview



Source: Kaggle+Synthetic data



Classes: A+, A-, AB+, AB-, B+, B-, O+, O-



Samples: Train (2800), Validation (600), Test (600)

Dataset Overview	Details
Number of Classes	8(A+, A-, AB+, AB-, B+,B-,O+,O-)
Training Samples	2800
Validation Samples	600
Test Samples	600

✓ **Dataset Size: $500 \times 8 = 4000$**

✓ **Split : 70% Train, 15% Validation, 15% Test**

Set	Images per class	Total images
Train	350	2800
Validation	75	600
Test	75	600

Subset Classification Based on Temperature Range

Subset Category	Temperature Range (°C)	Description
Low	Below 36°C	Indicates cooler blood sample images
Normal	36°C – 37.5°C	Represents typical physiological conditions
High	Above 37.5°C	Simulates fever or heated sample conditions

```
PS D:\pythonimageprocessing> & D:/pythonimageprocessing/blood_group_env/Scripts/python.exe d:/pythonimageprocessing/split_data.py
```

- ✓ Split A Nageative/high: 117 train, 25 validation, 26 test
- ✓ Split A Nageative/low: 116 train, 25 validation, 25 test
- ✓ Split A Nageative/normal: 116 train, 25 validation, 25 test
- ✓ Split A Positive/high: 117 train, 25 validation, 26 test
- ✓ Split A Positive/low: 116 train, 25 validation, 25 test
- ✓ Split A Positive/normal: 116 train, 25 validation, 25 test
- ✓ Split AB Negative/high: 117 train, 25 validation, 26 test
- ✓ Split AB Negative/low: 116 train, 25 validation, 25 test
- ✓ Split AB Negative/normal: 116 train, 25 validation, 25 test
- ✓ Split AB Positive/high: 117 train, 25 validation, 26 test
- ✓ Split AB Positive/low: 116 train, 25 validation, 25 test
- ✓ Split AB Positive/normal: 116 train, 25 validation, 25 test
- ✓ Split B Negative/high: 117 train, 25 validation, 26 test
- ✓ Split B Negative/low: 116 train, 25 validation, 25 test
- 25 validation, 25 test
- ✓ Split O Negative/high: 117 train, 25 validation, 26 test
- ✓ Split O Negative/low: 116 train, 25 validation, 25 test
- 25 validation, 25 test
- ✓ Split O Negative/high: 117 train, 25 validation, 26 test
- ✓ Split O Negative/low: 116 train, 25 validation, 25 test
- ✓ Split O Negative/normal: 116 train, 25 validation, 25 test
- ✓ Split O Positive/high: 117 train, 25 validation, 26 test
- ✓ Split O Positive/low: 116 train, 25 validation, 25 test
- ✓ Split O Positive/normal: 116 train, 25 validation, 25 test
- 🎯 Dataset split completed successfully!

```
PS D:\pythonimageprocessing>
```

```
PS D:\pythonimageprocessing> & D:/pythonimageprocessing/blood_group_env/Scripts/python.exe d:/pythonimageprocessing/preprocess.py
```

- ✓ Image preprocessing completed successfully!

10. Preprocessing Steps

1

Normalization

Images are normalized to a common scale for consistent processing.

2

Resizing

Images are resized to a uniform size for model input.

3

Data Augmentation

Techniques like rotation and zoom are applied to increase dataset diversity.

11. Model Architecture

Base: Pretrained VGG16

Custom Layers: GlobalAveragePooling2D, Dense, Softmax
Softmax

Classification: blood group classes
classes

12. Training Parameters

1 Epochs: 25

2 Batch Size: 16

3 Loss Function: Categorical
CrossEntropy (Loss: 0.39)

Output Screenshot

```
12 if train_acc is not None:
PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [ ] ...
174/174 24s 137ms/step - accuracy: 0.8750 - loss: 0.2464 - val_accuracy: 0.8733 - val_loss: 0.4457 - learning_rate: 1.0000e-04
Epoch 11/25
174/174 149s 858ms/step - accuracy: 0.8767 - loss: 0.4170 - val_accuracy: 0.8716 - val_loss: 0.4319 - learning_rate: 1.0000e-04
Epoch 12/25
174/174 24s 137ms/step - accuracy: 0.8125 - loss: 0.5416 - val_accuracy: 0.8716 - val_loss: 0.4309 - learning_rate: 1.0000e-04
Epoch 13/25
174/174 211s 1s/step - accuracy: 0.8633 - loss: 0.4217 - val_accuracy: 0.8750 - val_loss: 0.4043 - learning_rate: 2.0000e-05
Epoch 14/25
174/174 33s 183ms/step - accuracy: 0.9375 - loss: 0.2829 - val_accuracy: 0.8784 - val_loss: 0.4018 - learning_rate: 2.0000e-05
Epoch 15/25
174/174 145s 834ms/step - accuracy: 0.8717 - loss: 0.3938 - val_accuracy: 0.8733 - val_loss: 0.4050 - learning_rate: 2.0000e-05
Epoch 16/25
174/174 24s 133ms/step - accuracy: 0.8750 - loss: 0.5312 - val_accuracy: 0.8767 - val_loss: 0.3977 - learning_rate: 2.0000e-05
Epoch 17/25
174/174 144s 826ms/step - accuracy: 0.8666 - loss: 0.4009 - val_accuracy: 0.8733 - val_loss: 0.3973 - learning_rate: 2.0000e-05
Epoch 18/25
174/174 24s 132ms/step - accuracy: 1.0000 - loss: 0.1797 - val_accuracy: 0.8750 - val_loss: 0.3927 - learning_rate: 2.0000e-05
Epoch 19/25
174/174 146s 838ms/step - accuracy: 0.8608 - loss: 0.4036 - val_accuracy: 0.8733 - val_loss: 0.4032 - learning_rate: 2.0000e-05
Epoch 20/25
174/174 23s 131ms/step - accuracy: 1.0000 - loss: 0.0903 - val_accuracy: 0.8750 - val_loss: 0.4009 - learning_rate: 2.0000e-05
Epoch 21/25
```

```
PS D:\pythonimageprocessing> & D:/pythonimageprocessing/blood_group_env/Scripts/python.exe d:/pythonimageprocessing/training_validation_accuracy_heck.py
```

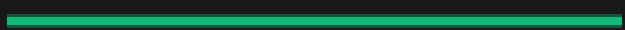
✓ Final Training Accuracy: 87.25%

✓ Final Validation Accuracy: 86.99%

i Test accuracy not found in history. Check your evaluation script separately.

```
PS D:\pythonimageprocessing>
```

```
self._warn_if_super_not_called()
```

```
608/608  56s 91ms/step - accuracy: 0.9662 - loss: 0.2243
```

 Total Test Samples: 608

 Number of Classes: 8

 Test Accuracy: 87.34%, Test Loss: 0.4320

```
PS D:\pythonimageprocessing>
```


13. Evaluation Metrics



Training Accuracy:
87.36%



Validation Accuracy:
86.83%

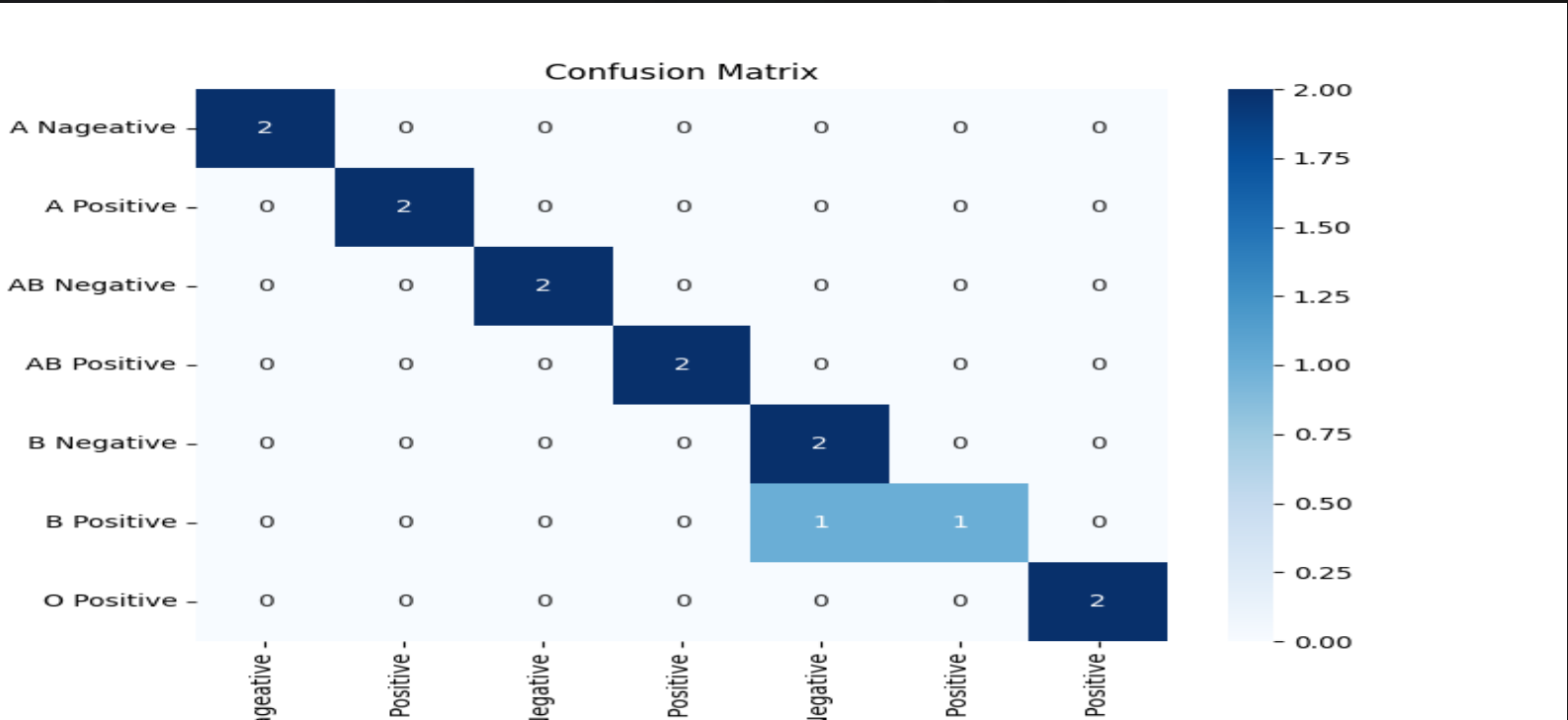
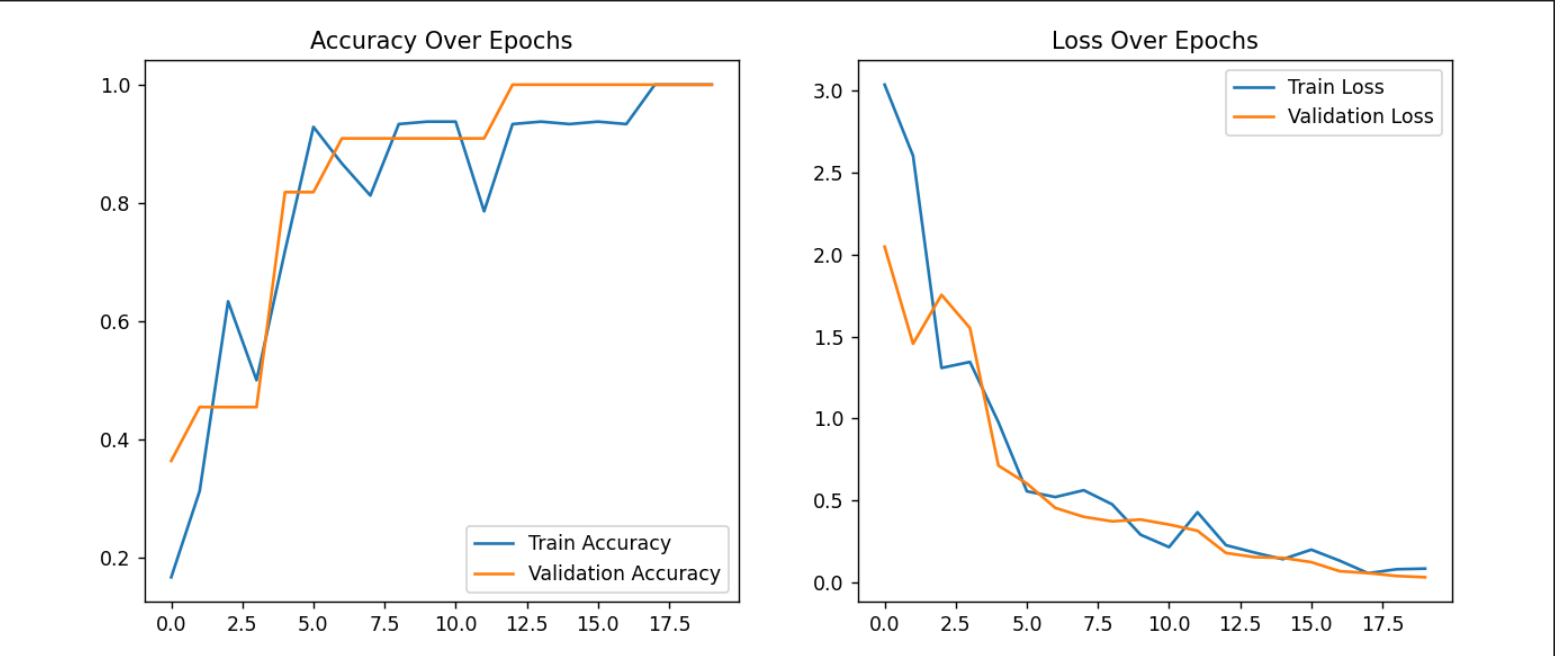


Test Accuracy:
87.34%

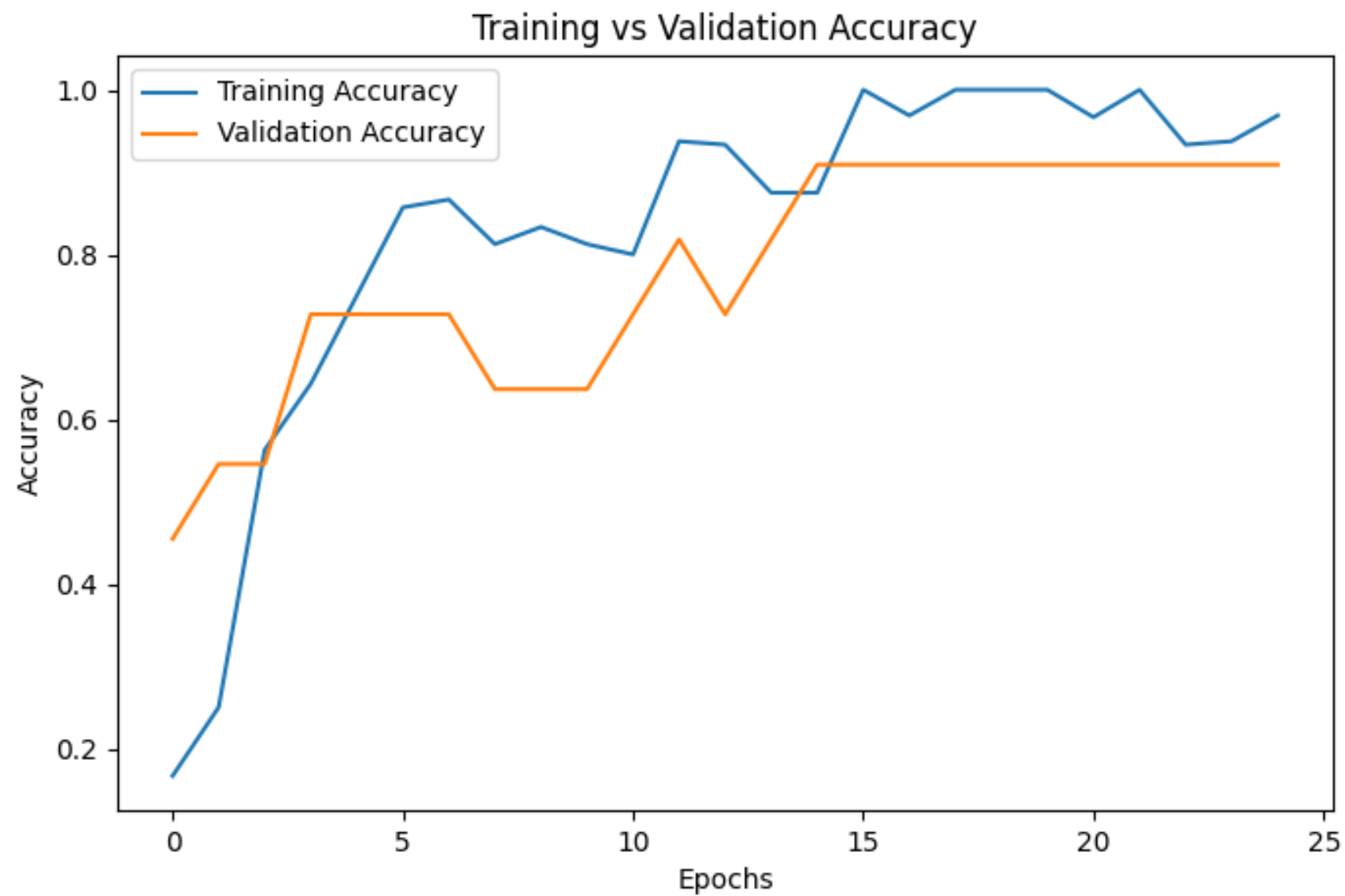


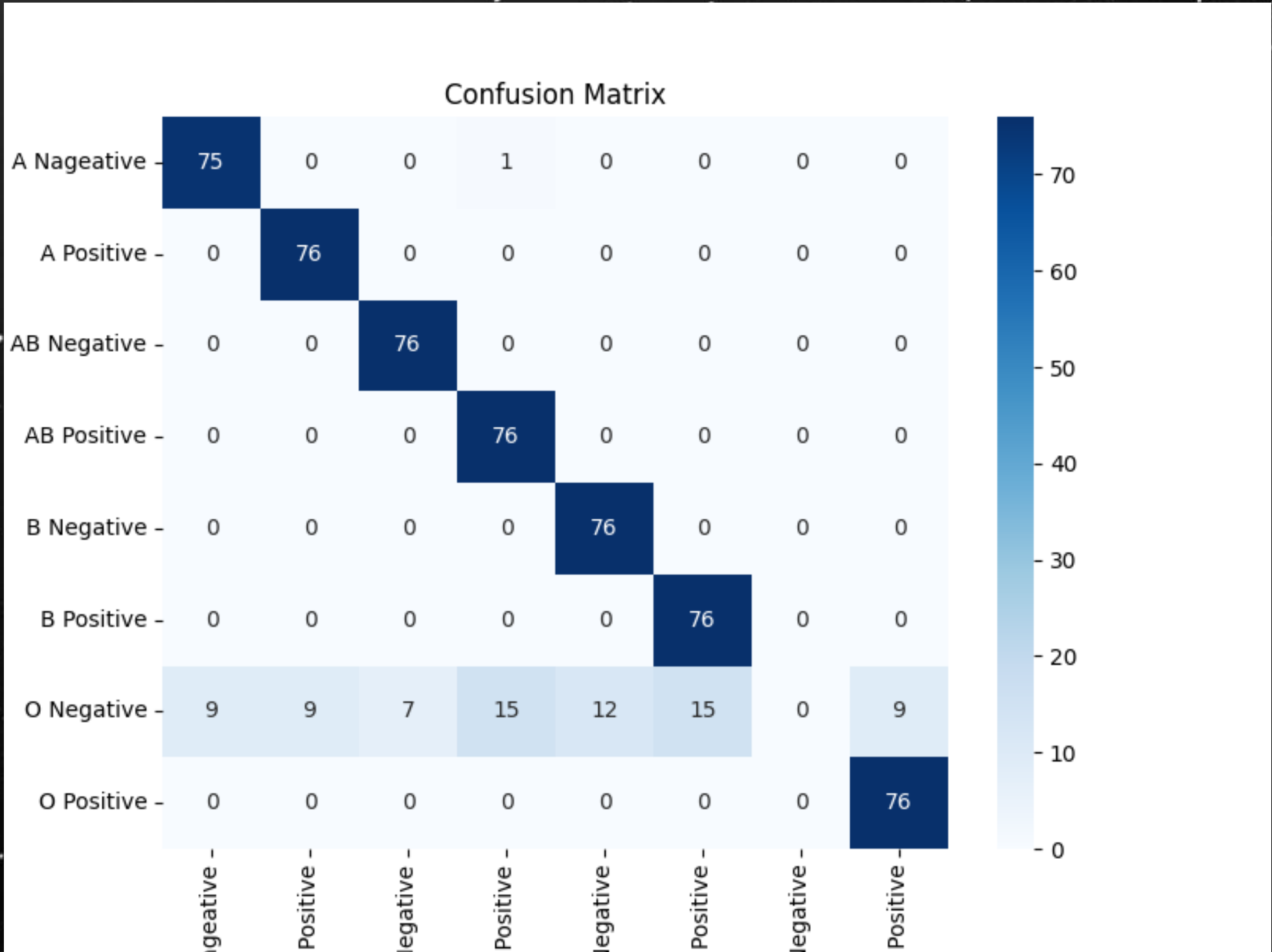
Metrics Used: Accuracy,
Precision, Recall, F1
Score

14. Confusion Matrix



14. Updated Confusion Matrix







15. MVP – Web Application Demo

- 1 Flask-based app for real-time predictions
- 2 Upload image and Temp. → Get result
- 3 Demo screenshots / small GIF preview

User Interface

Blood Group Detection

Upload Infrared Image:

No file chosen

Enter Temperature (°C):

Predict

Blood Group Detection

Upload Infrared Image:

AB_Positive_9.jpg

Enter Temperature (°C):

37

Predict



Prediction Successful

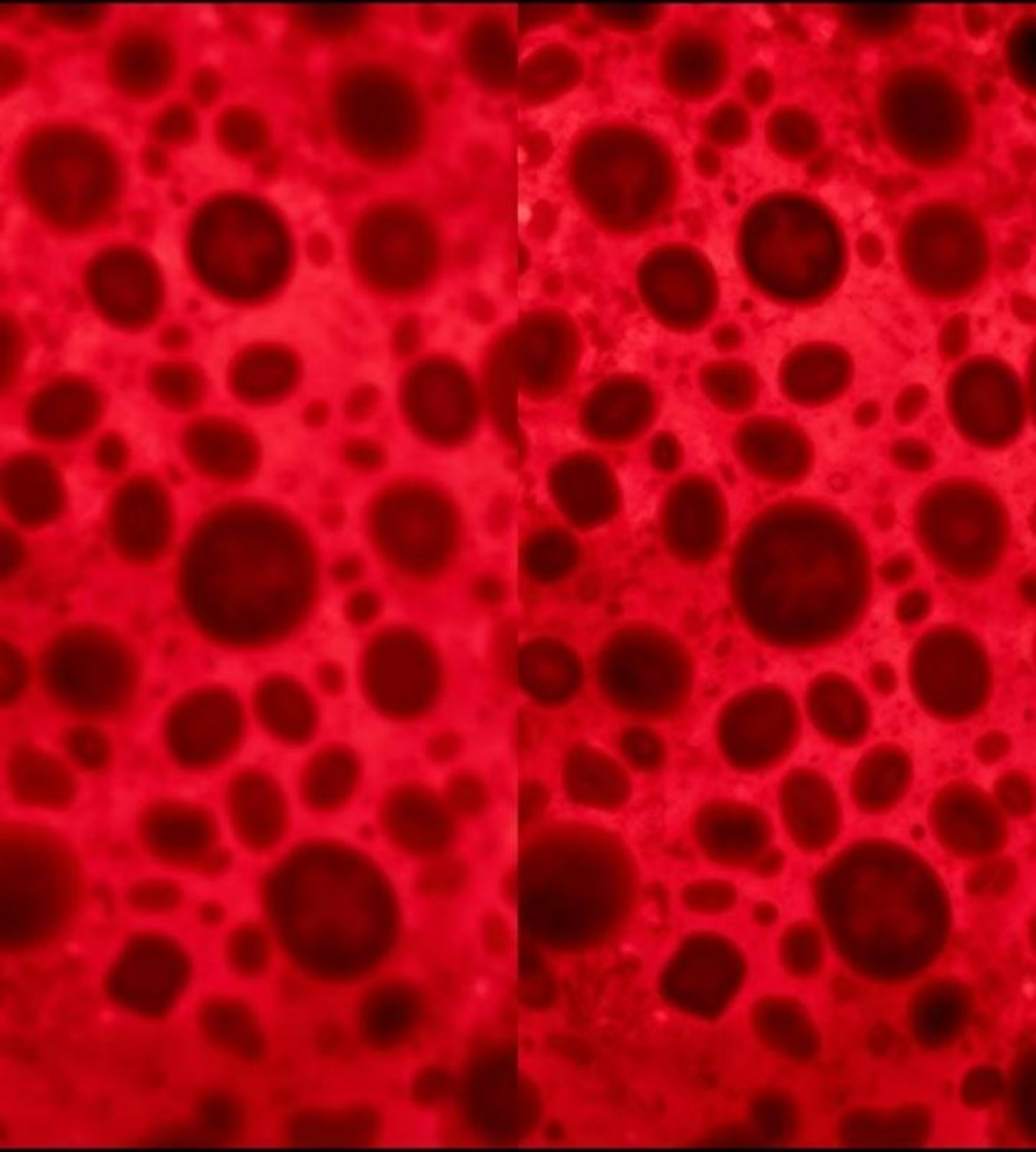
Predicted Blood Group: AB Positive

Entered Temperature: 37.0 °C

Predict Another

16. Go-To-Market Strategy

- Target hospitals, labs, and researchers.
- Promote via SEO, social media, and email outreach.
- Offer freemium web app to drive adoption.
- Partner with med-tech firms and hospital IT providers.
- Attend medical expos and tech conferences.
- Share tutorials and case studies online.
- Gather feedback to improve and retain users.

A vertical strip on the left side of the slide shows two panels of InfraRed Spectroscopic Imaging. The panels display a dense field of cells, with each cell appearing as a bright, circular spot against a darker background. The cells vary in size and intensity, suggesting different internal structures or compositions.

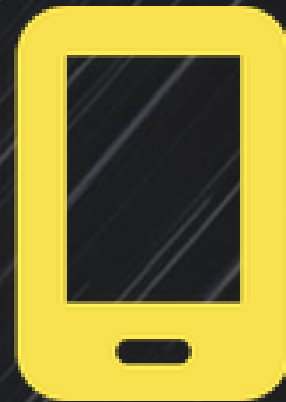
17. Uniqueness & Innovation

- 1 InfraRed Spectroscopic Imaging (vs. traditional agglutination)
- 2 ML-Powered Prediction – 87.34% accurate
- 3 Portable, Real-time Web App
- 4 No skilled personnel needed

18. Future Roadmap



Add O-, rare blood groups



Deploy as mobile app

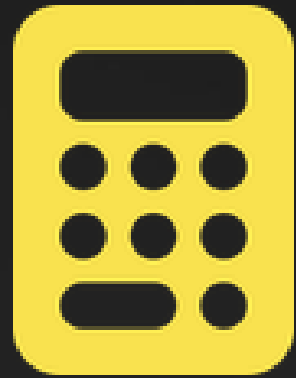


Integrate with hospital databases



Hardware prototype with IR camera

19. Conclusion & Final Ask



Successfully built an
ML system with
87.34% test accuracy



Real-time web
interface ready



Support for field testing,
scaling, mentorship, and
funding